

STANDING ON THE SHOULDERS OF

CELESTIALS

15 Samples vs. 1.76 Trillion Parameters

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The design process IS algorithmic, and recognizing this transforms designers from algorithm consumers into algorithm creators.

The LinnDrum didn't replace drummers. It taught them to think computationally. Roger Linn designed the algorithm by setting parameters: 15 drum samples, specific attack times,



and carefully tuned decay. Parameters aren't constraints. They're the design. ChatGPT feels scary because its newest model is estimated to have 1.76 trillion parameters. Estimated. Big Tech won't tell us the actual number, or they've lost track themselves. Now that is scary.

This opacity is intentional. Big Tech wants designers to believe algorithms are too complex. But designers have always been computational thinkers; we've just been trained to forget it.

Algorithmic Anxiety

When you map a user flow, you're writing an algorithm. When you define interaction states, you're setting parameters. When you decide what users see and when, you're making the same choices Roger Linn made with his 15 samples. The difference? Linn defined his parameters. Today's designers are lost somewhere in the middle.

It's only natural for designers to feel a sense of algorithmic anxiety. ChatGPT feels like this black void that has unlimited ideas, never gets tired, and always stays positive, because it's programmed that way. While there are dangers with this lack of autonomy on what AI is doing I believe the best thing a designer can do is go into this "black void" with intentionality.

Tell it to make an SVG pattern generator, and BOOM the HTML is open

LinnDrum

Creativity requires human experience. Algorithms require design.

The LM-1 contained 15 hand-sampled acoustic drum sounds. Fifteen. At \$5,500, only 500 were ever manufactured—not for exclusivity, but because they were expensive to produce. Yet Linn offered something no one else did: send him a tape, get custom chips back. Your sounds. Your parameters. While competitors raced to add more presets, Linn built control. His 15 samples were curated for musicality, for feel.

But he didn't stop there, he let musicians customize. Prince shaped his parameters. Michael Jackson shaped his. The algorithm served the artist because the artist was embedded in the design process. That was computational thinking. This is what designers do: they make computational choices. Fifteen samples vs. 1.76 trillion parameters. Linn's 15 were enough because they were designed, not just computed.

More parameters doesn't mean better results, it often means less control. Quality of parameters

The Shift

When designers recognize the design process IS algorithmic, they stop being controlled by algorithms and start creating them. ChatGPT has 1.76 trillion parameters you can't see. So set the ones you can.

Define constraints. Specify structure. Control outputs.

You're not asking questions, you're rewriting instructions.



To Conclude

Computational thinking doesn't require coding. It requires recognizing that design decisions ARE computational. Master this, and the fun begins: customization, exploration, building your own algorithmic systems instead of decorating someone else's.

Whether Linn knew this theoretically or not, he mastered it in practice. Fifteen samples. That's all it took to shape a decade of music. Not 1.76 trillion parameters. Not infinite customization.

Fifteen curated, intentional choices. I'm not suggesting Roger Linn solved Big Tech's extraction problem by sampling acoustic drums in 1980. But he demonstrated something more urgent: designers don't need to learn computational thinking. We already do it. Every user flow is an algorithm. Every interaction state is a parameter.

Every design decision is computational. The anxiety dissolves when you recognize this. Big Tech wants you to forget that algorithms are designed by people making choices. Those fifteen samples? Linn's reminder that we've always had the agency. We just stopped claiming it.

